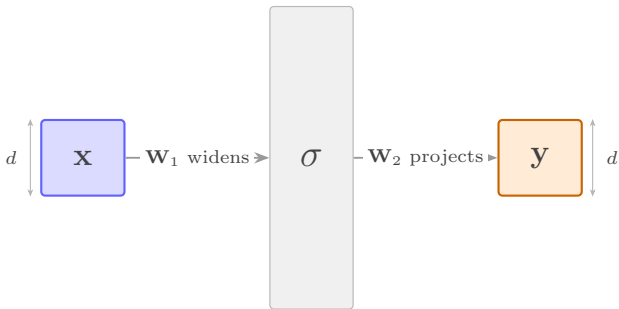


Position-wise feed-forward network

$$d_{\text{ff}} = 4d$$



$\text{FFN}(\mathbf{x}) = \sigma(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2$ · applied to each token independently
widen $d \rightarrow d_{\text{ff}}$, bend by σ (ReLU / GELU), project $d_{\text{ff}} \rightarrow d$ (typically $d_{\text{ff}} = 4d$)